

23AEE01 : Basic Electrical and Electronics Engineering

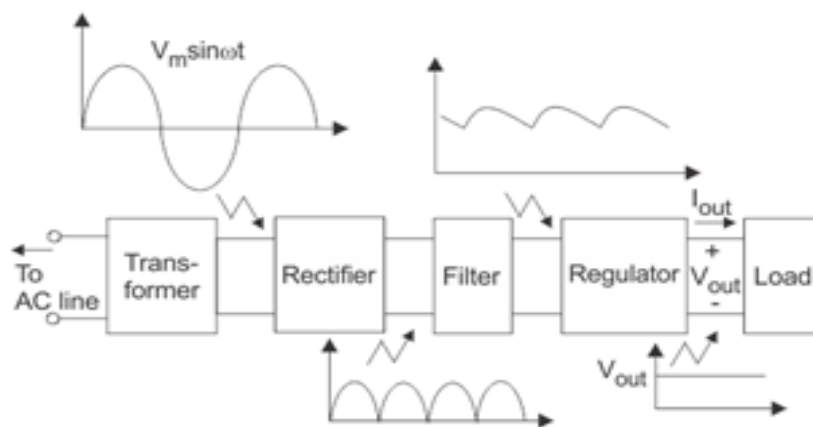
Part B : Basic Electronics Engineering

UNIT – II BASIC ELECTRONIC CIRCUITS AND INSTRUMENTATION

Power Supplies : Block diagram description of a DC power Supply:

A regulated power supply can convert unregulated an AC (alternating current or voltage) to a constant DC (direct current or voltage). A regulated power supply is used to ensure that the output remains constant even if the input changes. A regulated DC power supply is also called as a linear power supply; it is an embedded circuit and consists of various blocks.

The regulated power supply will accept an AC input and give a constant DC output. Figure below shows the block diagram of a typical regulated DC power supply



Components of typical linear power supply

The basic building blocks of a regulated DC power supply are as follows:

1. A step down transformer
2. A rectifier
3. A DC filter
4. A regulator

Step Down Transformer

A step down transformer will step down the voltage from the ac mains to the required voltage level. The turn's ratio of the transformer is so adjusted such as to obtain the required voltage value. The output of the transformer is given as an input to the rectifier circuit.

Rectification

Rectifier is an electronic circuit consisting of diodes which carries out the rectification process. Rectification is the process of converting an alternating voltage or current into corresponding direct (DC) quantity. The input to a rectifier is ac whereas its output is unidirectional pulsating DC. Usually a full wave rectifier or a bridge rectifier is used to rectify both the half cycles of the ac supply (full wave rectification). Figure below shows a full wave bridge rectifier.'

DC Filtration

The rectified voltage from the rectifier is a pulsating DC voltage having very high ripple content. But this is not we want, we want a pure ripple free DC

Regulation

This is the last block in a regulated DC power supply. The output voltage or current will change or fluctuate when there is change in the input from ac mains or due to change in load current at the output of the regulated power supply or due to other factors like temperature changes. This problem can be eliminated by using a regulator. A regulator will maintain the output constant even when changes at the input or any other changes occur. Transistor series regulator, Fixed and variable IC regulators or a zener diode operated in the zener region can be used depending on their applications. IC's like 78XX and 79XX are used to obtain fixed values of voltages at the output waveform. Hence a filter is used. Different types of filters are used such as capacitor filter, LC filter, Choke input filter, π type filter.

Rectifiers:

A rectifier is an electrical component that converts alternating current (AC) to direct current (DC). A rectifier is analogous to a one-way valve that allows an electrical current to flow in only one direction. The process of converting AC current to DC current is known as rectification. A rectifier can take several physical forms, such as solid-state diodes, vacuum tube diodes, mercury-arc valves, silicon-controlled rectifiers, and various other silicon-based semiconductor switches

Applications and Uses of Rectifiers

The primary application of the rectifier is to derive DC power from AC power. Rectifiers are used inside the power supplies of almost all electronic equipment. In power supplies, the rectifier is normally placed in series following the transformer, a smoothing filter, and possibly a voltage regulator

Half-wave rectifiers

Half-wave rectifiers transform AC voltage to DC voltage. A half-wave rectifier circuit uses only one diode for the transformation. A half-wave rectifier is defined as a type of rectifier that allows only one-half cycle of an AC voltage waveform to pass while blocking the other half cycle

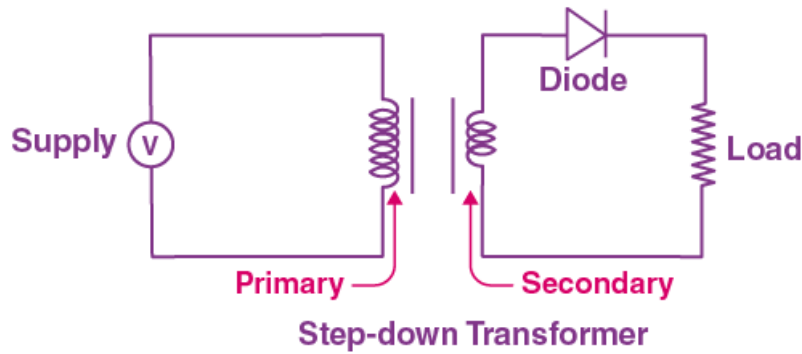
Half Wave Rectifier Circuit

A half-wave rectifier is the simplest form of the rectifier and requires only one diode for the construction of a half-wave rectifier circuit.

A half-wave rectifier circuit consists of three main components as follows:

- A diode
- A transformer
- A resistive load

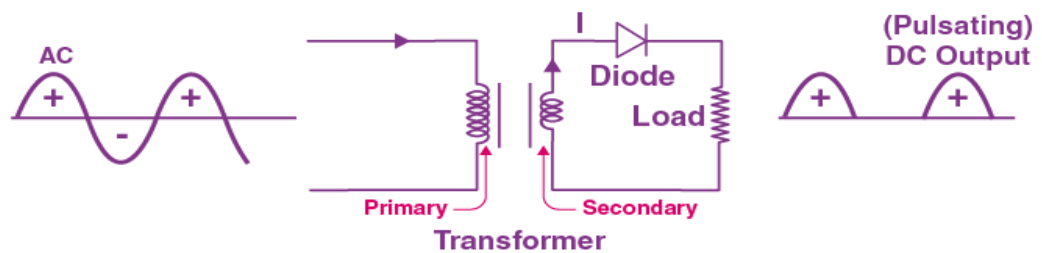
Given below is the half-wave rectifier diagram:



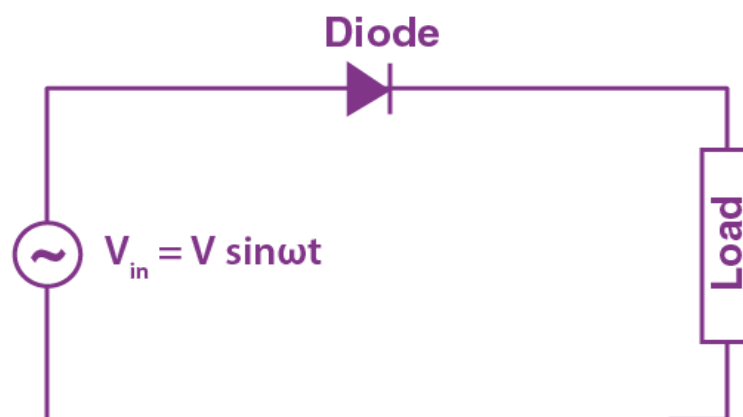
Working of Half Wave Rectifier

In this section, let us understand how a half-wave rectifier transforms AC into DC.

1. A high AC voltage is applied to the primary side of the step-down transformer. The obtained secondary low voltage is applied to the diode.
2. The diode is forward biased during the positive half cycle of the AC voltage and reverse biased during the negative half cycle.
3. The final output voltage waveform is as shown in the figure below:



For better understanding, let us simplify the half-wave circuit by replacing the secondary transformer coils with a voltage source as shown below:



For the positive half cycle of the AC source voltage, the circuit effectively becomes as shown below in the diagram:



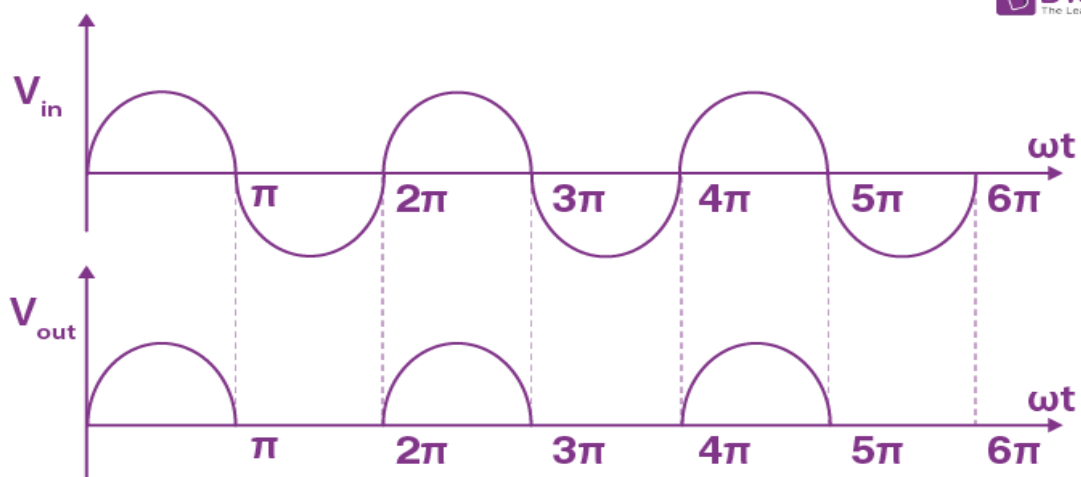
When the diode is forward biased, it acts as a closed switch. But, during the negative half cycle of the AC source voltage, the equivalent circuit becomes as shown in the figure below



when a diode is reverse biased, it acts as an open switch. Since no current can flow to the load, the output voltage is equal to zero.

Half Wave Rectifier Waveform

The half-wave rectifier waveform before and after rectification is shown below in the figure.



Applications of Half Wave Rectifier

Here are a few common applications of half wave rectifiers:

- They are used for signal demodulation purpose
- They are used for rectification applications
- They are used for signal peak applications

Disadvantages of Half Wave Rectifier

- Power loss
- Low output voltage
- The output contains a lot of ripples
-

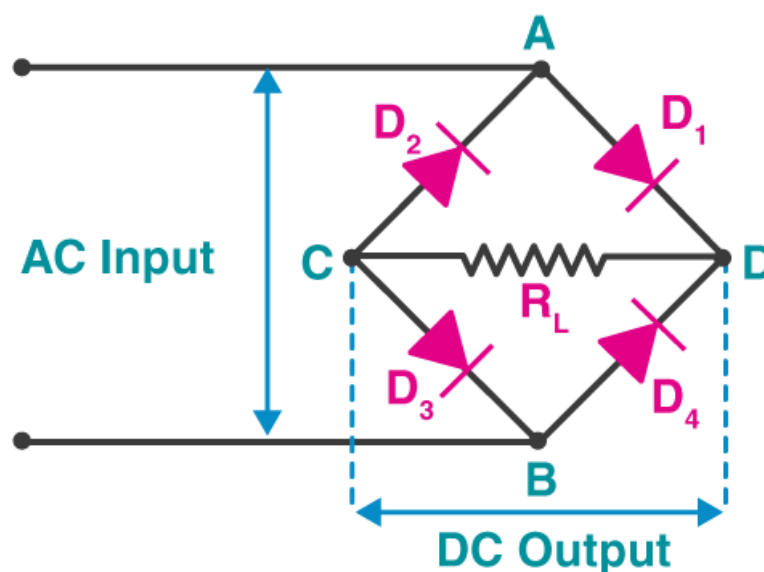
Full Wave Bridge Rectifier:

Working of a Full Wave Bridge Rectifier:

We can define bridge rectifiers as a type of full-wave rectifier that uses four or more diodes in a bridge circuit configuration to efficiently convert alternating (AC) current to a direct (DC) current.

Construction

The construction of a bridge rectifier is shown in the figure below. The bridge rectifier circuit is made of four diodes D_1 , D_2 , D_3 , D_4 , and a load resistor R_L . The four diodes are connected in a closed-loop configuration to efficiently convert the alternating current (AC) into Direct Current (DC). The main advantage of this configuration is the absence of the expensive centre-tapped transformer. Therefore, the size and cost are reduced.



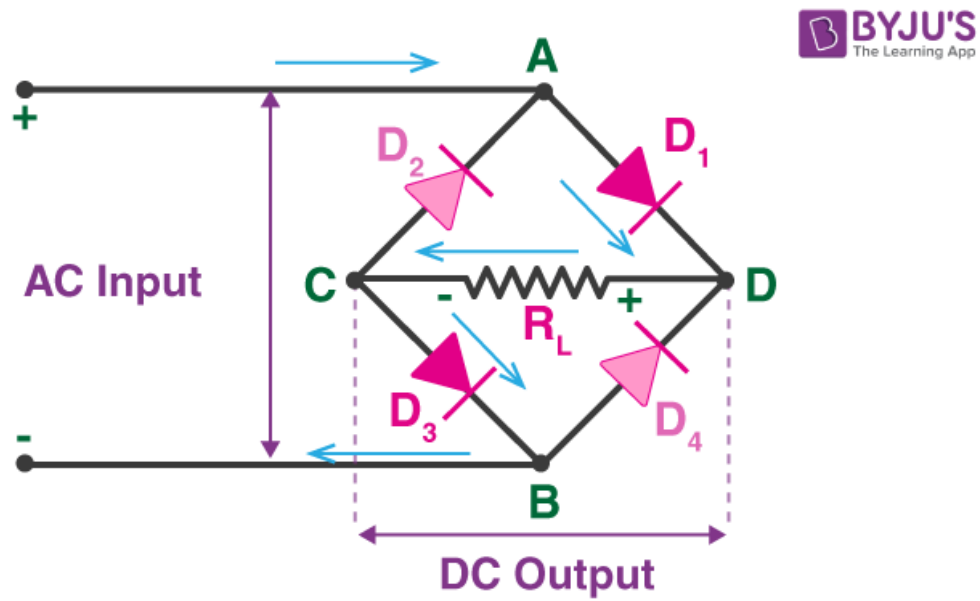
The input signal is applied across terminals A and B, and the output DC signal is obtained across the load resistor R_L connected between terminals C and D. The four diodes are arranged in such a way that only two diodes conduct electricity during each half cycle.

D_1 and D_3 are pairs that conduct electric current during the positive half cycle/. Likewise, diodes D_2 and D_4 conduct electric current during a negative half cycle.

Working

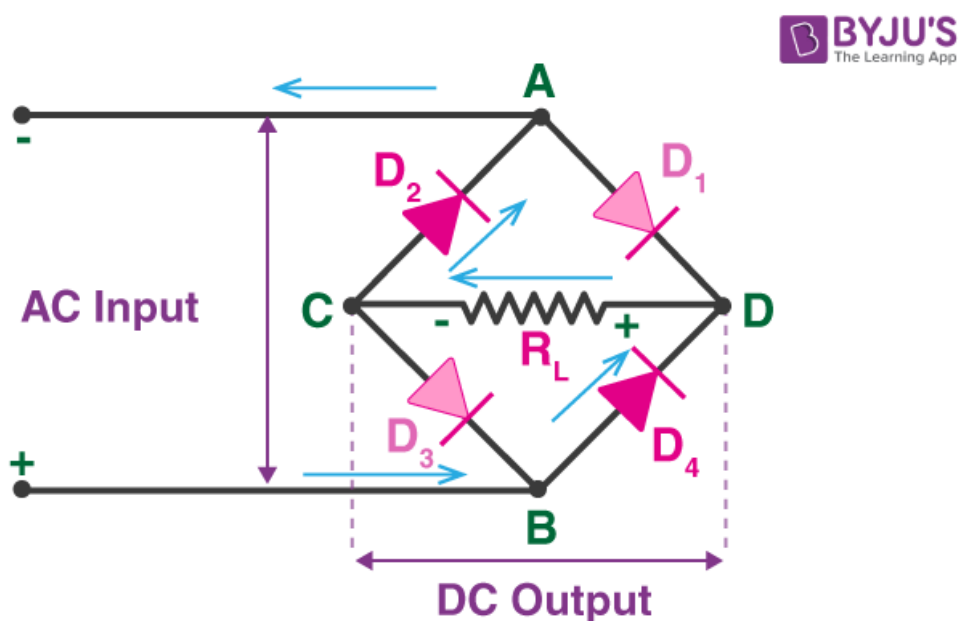
When an AC signal is applied across the bridge rectifier, terminal A becomes positive during the positive half cycle while terminal B becomes negative. This results in diodes D_1 and D_3 becoming forward biased while D_2 and D_4 becoming reverse biased.

The current flow during the positive half-cycle is shown in the figure below:



During the negative half-cycle, terminal B becomes positive while terminal A becomes negative. This causes diodes D_2 and D_4 to become forward biased and diode D_1 and D_3 to be reverse biased.

The current flow during the negative half cycle is shown in the figure below:

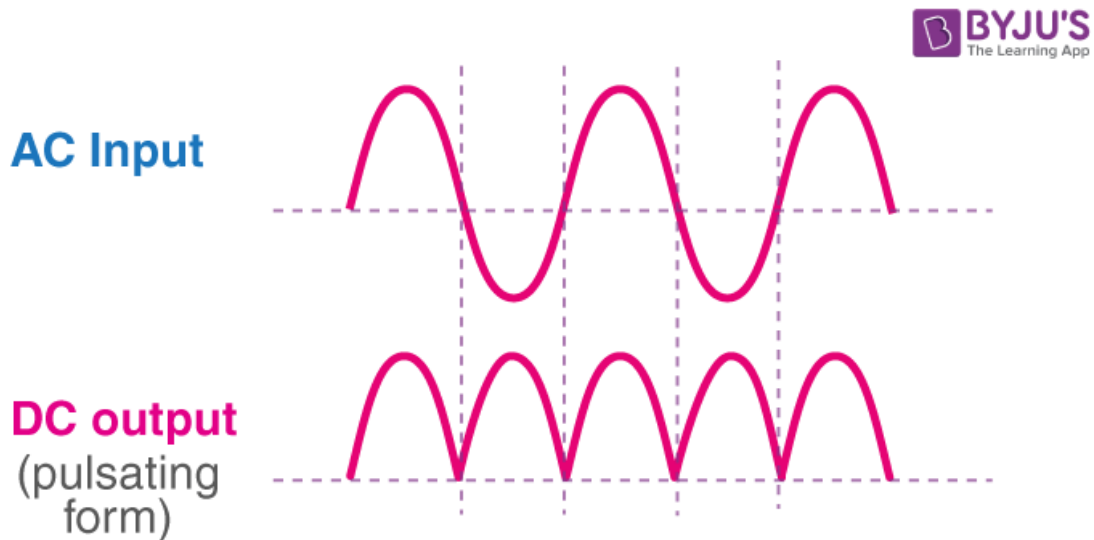


From the figures given above, we notice that the current flow across load resistor R_L is the same during the positive and negative half-cycles. The output DC signal polarity may be

either completely positive or negative. In our case, it is completely positive. If the diodes' direction is reversed, we get a complete negative DC voltage.

Thus, a bridge rectifier allows electric current during both positive and negative half cycles of the input AC signal.

The output waveforms of the bridge rectifier are shown in the below figure.



Characteristics of Bridge Rectifier

Ripple Factor

The smoothness of the output DC signal is measured by a factor known as the ripple factor. The output DC signal with fewer ripples is considered a smooth DC signal while the output with high ripples is considered a high pulsating DC signal.

Mathematically, the ripple factor is defined as the ratio of ripple voltage to pure DC voltage.

For bridge rectifiers, the ripple factor is 0.48.

Peak Inverse Voltage

The maximum voltage that a diode can withstand in the reverse bias condition is known as a peak inverse voltage. During the positive half cycle, the diodes D_1 and D_3 are in the conducting state while D_2 and D_4 are in the non-conducting state. Similarly, during the negative half cycle, diodes D_2 and D_4 are in the conducting state, and diodes D_1 and D_3 are in the non-conducting state.

Efficiency

The rectifier efficiency determines how efficiently the rectifier converts Alternating Current (AC) into Direct Current (DC). Rectifier efficiency is defined as the ratio of the DC output power to the AC input power. The maximum efficiency of a bridge rectifier is 81.2%.

Advantages

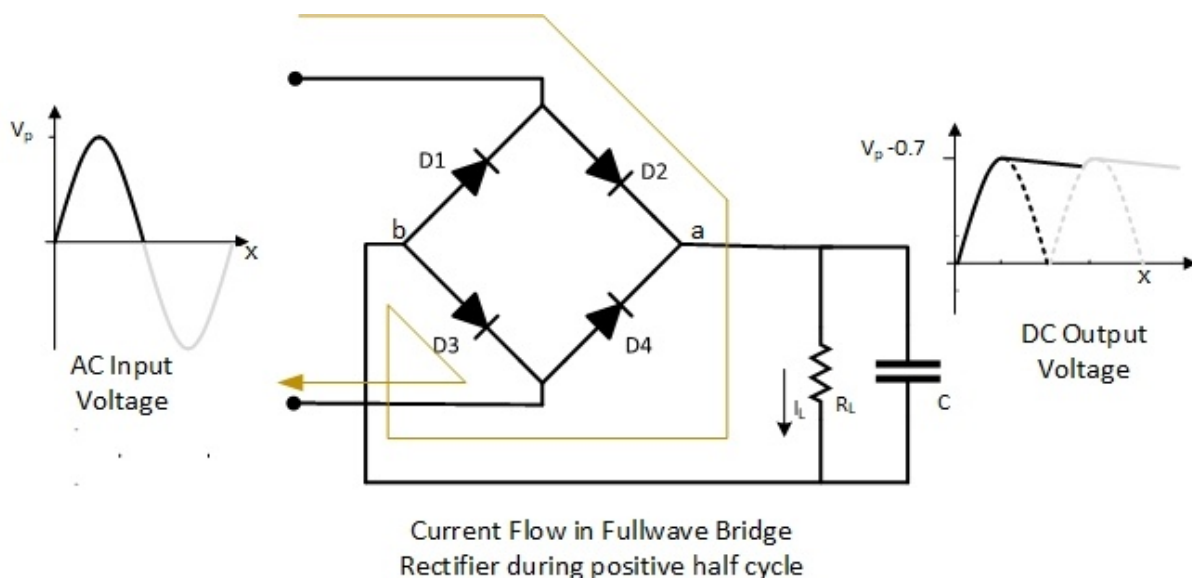
- The efficiency of the bridge rectifier is higher than the efficiency of a half-wave rectifier. However, the rectifier efficiency of the bridge rectifier and the centre-tapped full-wave rectifier is the same.
- The DC output signal of the bridge rectifier is smoother than the output DC signal of a half-wave rectifier.
- In a half-wave rectifier, only half of the input AC signal is used, and the other half is blocked. Half of the input signal is wasted in a half-wave rectifier. However, in a bridge rectifier, the electric current is allowed during both positive and negative half cycles of the input AC signal. Hence, the output DC signal is almost equal to the input AC signal.

Disadvantages

- The circuit of a bridge rectifier is complex when compared to a half-wave rectifier and centre-tapped full-wave rectifier. Bridge rectifiers use 4 diodes while half-wave rectifiers and centre-tapped full wave rectifiers use only two diodes.
- When more diodes are used more power loss occurs. In a centre-tapped full-wave rectifier, only one diode conducts during each half cycle. But in a bridge rectifier, two diodes connected in series conduct during each half cycle. Hence, the voltage drop is higher in a bridge rectifier.

Full Wave Bridge Rectifier with Capacitor Filter

The average output of the bridge rectifier is about 64% of the input voltage. The Bridge type full wave rectifier can convert an AC to DC by the mean of four diodes. The diodes are connected in such a configuration that the output peak voltage remains equal to the secondary of the transformer peak. In each half-cycle, a set of two diodes conduct and block the current alternately. In contrast to the center-tapped rectifier, the Bridge rectifier needs four diodes instead of two diodes, which becomes expensive.



Conclusion :

- The bridge rectifier converts both halves of the AC input cycle into DC output
- The rectifier uses four diodes that's why it is considered expensive
- The average output of the Bridge rectifier is twice that of a half-wave rectifier
- The ripple voltage of a full-wave bridge rectifier with a capacitor filter is less than that of a half-wave rectifier

Simple Zener Voltage Regulator: (Zener Diode as Voltage Regulator)

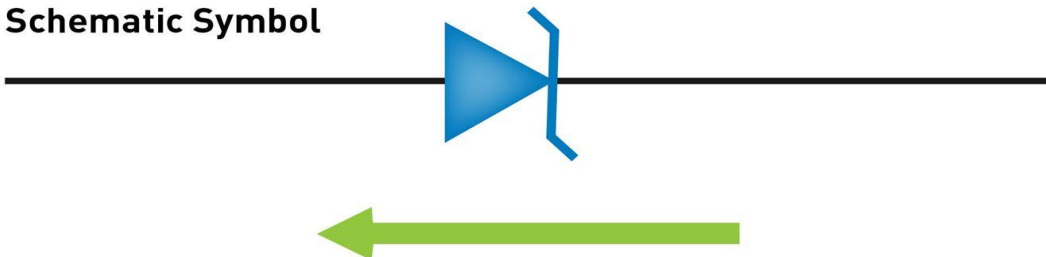
Zener diode is a silicon semiconductor with a p-n junction that is specifically designed to work in the reverse biased condition. When forward biased, it behaves like a normal signal diode, but when the reverse voltage is applied to it, the voltage remains constant for a wide range of currents. Due to this feature, it is used as a voltage regulator in d.c. circuit. The primary objective of the Zener diode as a voltage regulator is to maintain a constant voltage. Let us say if Zener voltage of 5 V is used then, the voltage becomes constant at 5 V, and it does not change.



Appearance



Schematic Symbol



Backwards current flow too, but only past the “Zener” breakdown voltage

A

Semiconductor Diode blocks current in the reverse direction but will suffer damage if the reverse voltage applied across becomes too high.

A voltage regulator is a device that regulates the voltage level. It essentially steps down the input voltage to the desired level and keeps it at that same level during the supply. This ensures that even when a load is applied the voltage doesn't drop. The voltage regulator is used for two main reasons, and they are:

- To vary or regulate the output voltage
- To keep the output voltage constant at the desired value in spite of variations in the supply voltage.

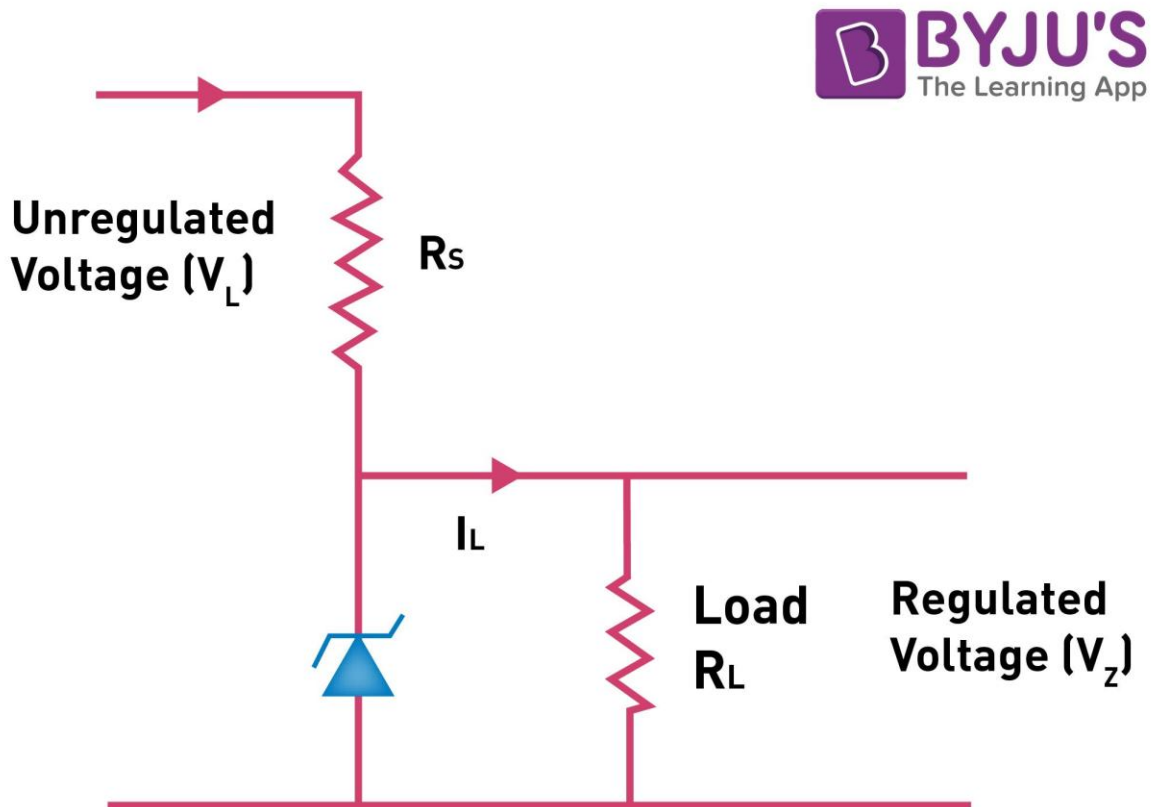
Voltage regulators are used in computers, power generators, alternators to control the output of the plant.

Zener Diode as a Voltage Regulator

There is a series resistor connected to the circuit in order to limit the current into the diode. It is connected to the positive terminal of the d.c. It works in such a way the reverse-biased can also work in breakdown conditions. We do not use ordinary junction diode because the low power rating diode can get damaged when we apply reverse bias above its breakdown voltage. When the minimum input voltage and the maximum load current is applied, the Zener diode current should always be minimum.

Since the input voltage and the required output voltage is known, it is easier to choose a Zener diode with a voltage approximately equal to the load voltage, i.e. $V_Z = V_L$.

The circuit diagram of a voltage regulator using a Zener diode is shown:



The value of the series resistor is written as $R_S = (V_L - V_Z)I_L$.

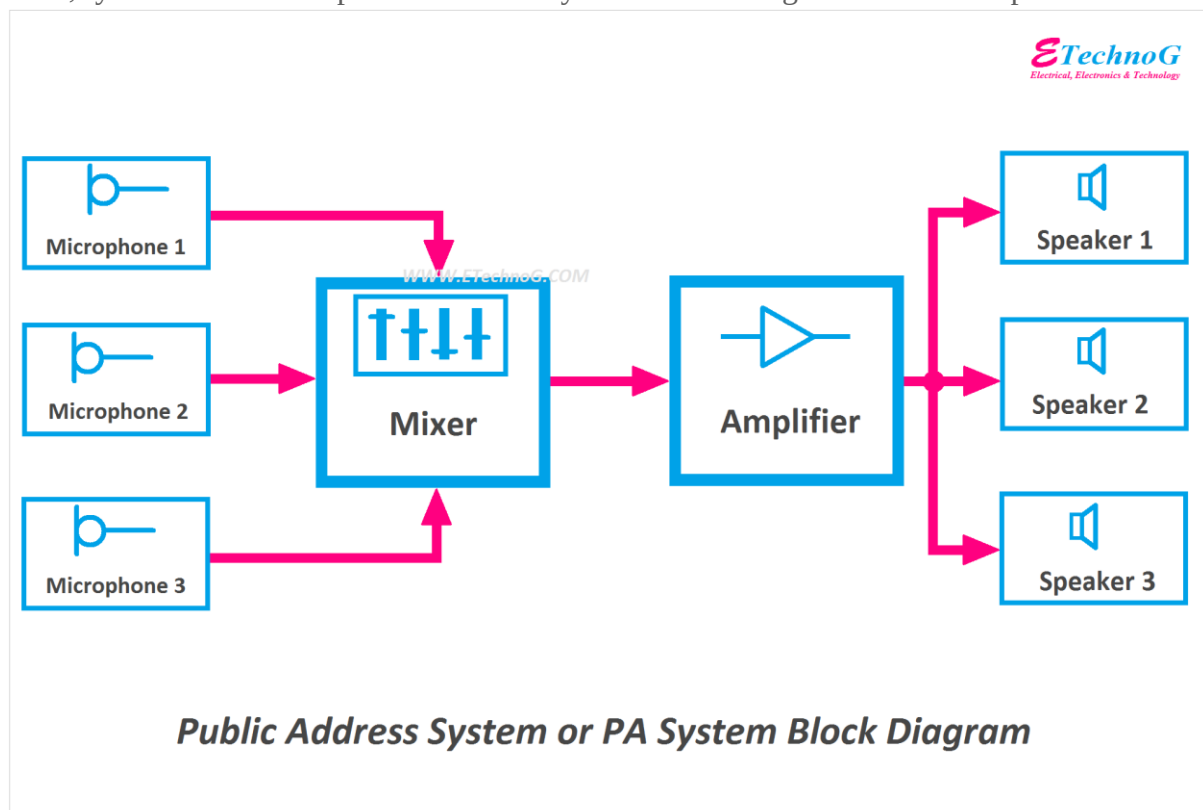
Current through the diode increases when the voltage across the diode tends to increase which results in the voltage drop across the resistor. Similarly, the current through the diode decreases when the voltage across the diode tends to decrease. Here, the voltage drop across the resistor is very less, and the output voltage results normally.

Amplifiers : Block Diagram of Public Address System

A public Address System or PA system is an electronic system that includes acoustic signal converting, mixing, amplifying, and playing. A PA system has microphones, amplifiers, and loudspeakers as its main components or equipment. We know that the intensity of the sound decreases with the distance. Also, a particular sound can be affected or distorted by the other sound if the level or volume of the sounds is the same. So, if we want to deliver the sound energy for a long-distance or highlight a particular sound or voice in a large crowd or gathering then we need a PA system. Because the PA systems provide the amplification of the sound for comfortable listening.

The range of amplification is required for the sound is to be delivered to the distance, size of the gathering or gathering, etc. Sound quality is also a very important factor with amplification because it improves the listening experience. The PA system helps to amplify as well as record the voice of any human being or the sound of any musical instrument. Also, it helps to communicate between groups.

Here, you can see the public address system block diagram and its important blocks.



Components of a PA System

There are so many devices or components are used in PA system that depends upon their applications and other factors. But the main three components of any public address system are explained below.

1. Microphone

The microphone is a very important part or component of a PA system. The microphone basically is a transducer that converts acoustic energy or sound energy into electrical energy. It continuously generates the pulsating electrical voltage according to the frequency of the sound energy applied to it. There various types of microphones are used in the PA system. The main basic two types of microphones are,

1. Handheld Microphone
2. Lapel Microphone

Other different types of microphones are,

- Wired Microphone

- **Wireless Microphone**

A wired microphone can be connected by a wire to the mixer or amplifier. It is very simple. But the wireless microphone needs a battery and the frequency of its signal is also a very important factor. The wireless microphone cannot be connected directly to the mixer or amplifier. A receiver is required that can receive the signals sent by the microphone. This receiver is to be connected to the amplifier.

2. Amplifier

The amplifier is the second part of a PA system. The main function of the amplifier is to amplify or increase the volume level of the audio signal that can drive a loudspeaker. The requirement or size of the amplifier depends upon the number and size of the loudspeaker. If the size of the loudspeaker is very large or so many loudspeakers are to be connected then a very powerful amplifier is required. The output of the amplifier is measured in watts or kilowatt.

3. LoudSpeaker

Loudspeaker provides a very important role in the PA system. It converts electrical energy into acoustic energy or sound energy. The loudspeakers are generally connected to the amplifier and it generates sound according to the audio signal provided by the amplifier. There are different types of loudspeakers available according to their operating frequency.

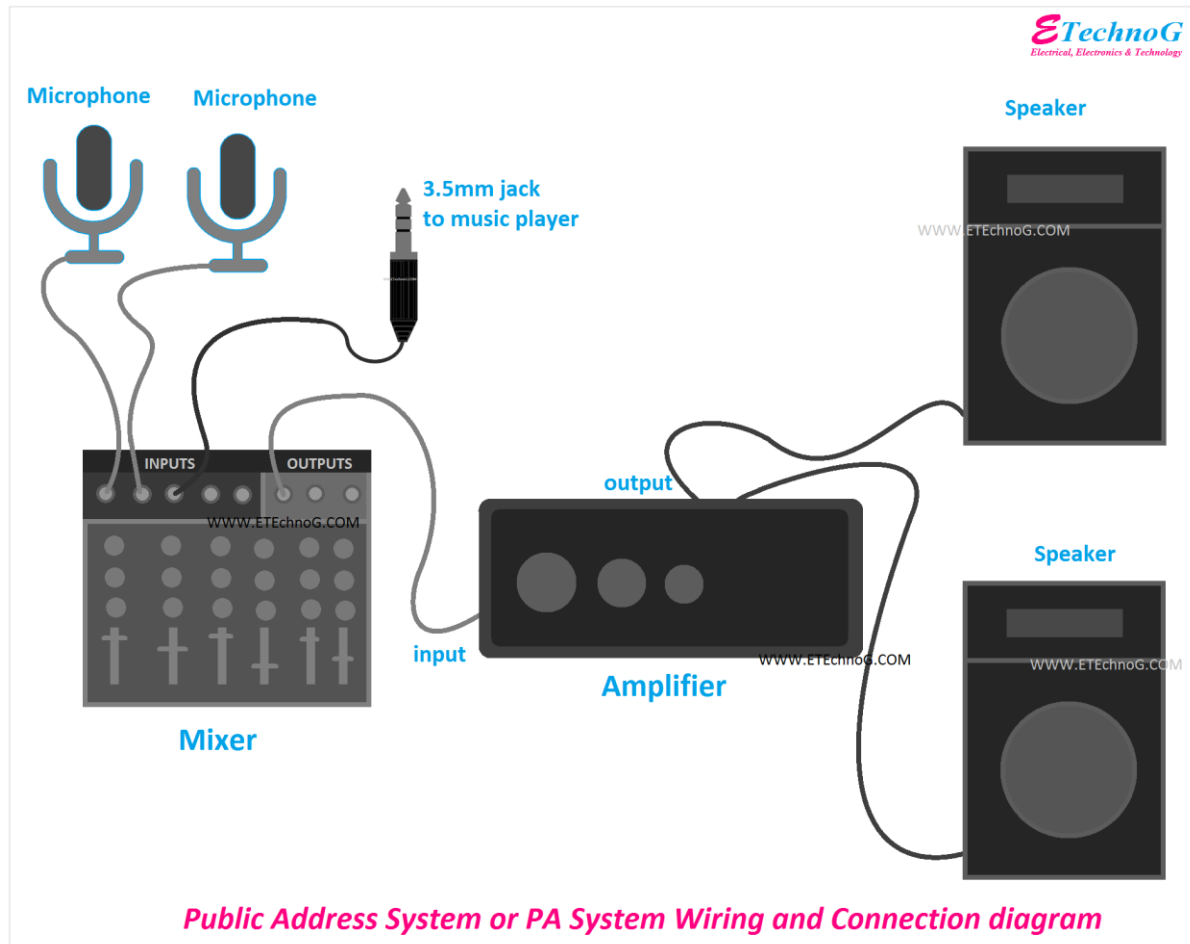
1. **Woofer** - it operates at the lowest frequency audio signal such as Bass
2. **Subwoofer** - it also operates with low frequency but more than the woofer such as bass and deep vocals
3. **Squawker** - it operates with medium-frequency audio signals such as vocals
4. **Tweeter** - it operates with high-frequency audio signals such as tone

4. Mixer

A mixer is not a necessary part of a PA system. But if there are multiple audio sources or multiple microphones used in the PA system then a mixer is must required. The mixer is an electronic device that can control multiple sound sources simultaneously. It can mix all the sound sources together and play with a single loudspeaker with the help of an amplifier. So the mixer is generally connected before the amplifier. Nowadays, there are so many amplifiers available in the market that already has an inbuilt mixer. So there is no need for an external additional mixer. All the microphones or sound sources can be connected directly to the amplifier.

PA System Wiring and Connection Diagram

Here, you can see a basic wiring diagram of public address system. This will help you understand how microphones, mixers, amplifiers, and speakers are connected together.



Here, you can see, the multiple audio or sound sources such as microphones and music players are connected to the mixer. Then the mixer output is connected to the amplifier input. And the amplifier output is connected to the speaker.

Important Factors in a Public Address System

Acoustic Feedback

When the microphone picks up sound from the speaker and re-amplifies and sent again to the speaker then it is called Acoustic feedback. The objective of any pa system should be to minimize as much as possible. It distorted the actual sound and make a bad experience of listening. It generally happens more when the volume of the system is turned up to the highest level.

This can be reduced by the different procedures,

- by keeping microphones at a distance from the speaker
- changing the direction of the microphone from the point towards the speaker
- lowering the gain level of the feedback sound
- using some devices like notch filter, graphic equalizer, parametric equalizer, etc

Cables and Wires

The size and quality of cables used in the PA system is a very important factor. Using the proper size and good quality cables and wire improves the sound quality very much.

Types of PA System

The PA system can be classified such as,

- Personal PA System
- Medium Size PA System
- Large Size PA System
- Portable PA System

Applications and Uses of PA System

1. PA system is used in group meetings, presentations, concerts, theaters, large halls, etc.
2. PA system also used in sports, stadiums, travel system, security system, conference system, etc.

Circuit Diagram and working of Common Emitter (RC Coupled) amplifier with its frequency response:

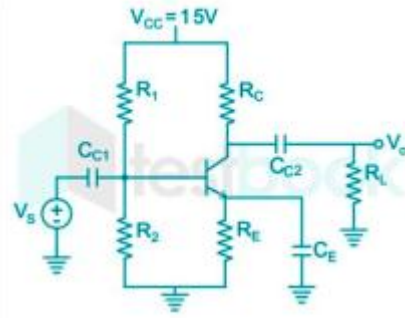
We know that a transistor is used as an amplifier, to obtain a high signal from a low signal quantity. The amplifier has to be designed in such a way it can amplify the signal and produce the output by varying the input applied to it. But for high gain and to reduce the loading effect we can use more than two amplifiers. By means of coupling this can be achieved, here we will discuss the concept of RC coupled amplifier and its operation.

RC coupled Amplifier

An RC-coupled amplifier is a type of electronic amplifier that uses a resistor-capacitor (RC) network to couple the output of one stage of amplification to the input of the next stage. The capacitor blocks any DC voltage, while allowing AC signals to pass through. The resistor and capacitor values are carefully selected to optimize the frequency response of the amplifier. This configuration is commonly used in audio amplifiers and radio receivers due to its simplicity and effectiveness.

What is RC coupled Amplifier?

A transistor circuit containing more than one stage of amplification is defined as a multistage transistor amplifier. In this amplifier, a number of single amplifier stages are connected in a **cascade arrangement**. Cascading means the output of the first stage is connected to the input of the second stage through a suitable coupling device and so on. The coupling devices are generally capacitors, transformers, etc.



Purpose of the RC coupled Amplifier device

The purpose of the coupling device is:

- To transfer a.c. the output of one stage, to the input of the next stage.
- To isolate the d.c. conditions of one stage from the next stage.
- To reduce or eliminate the loading effect.



Block Diagram of Multistage Amplifier

The above figure shows the block diagram of a 3 stage cascade amplifier. Each stage consists of a transistor amplifier and is coupled to the next stage through a coupling device. The commonly used cascade amplifiers are:

- R.C. coupled amplifier
- Transformer-coupled amplifier
- Direct-coupled amplifier

Construction of a Two stage RC Coupled Amplifier

The below figure shows a two-stage R.C. coupled amplifier. Here the first stage is connected to the second stage through the coupling capacitor CC and resistor RC. Therefore, this amplifier is called an RC-coupled amplifier. The resistors R1, R2 form the self-bias and RE form stabilization network. The bypass capacitor at the emitter offers a low reactance path to the signal. This method of coupling is widely used in cascade amplifier stages.

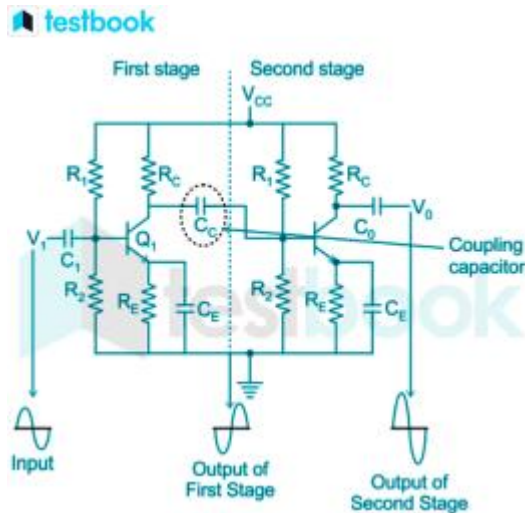
It can also drive low impedance loads because of its relatively low output impedance.

Due to the non-linear properties of the transistors, this sort of amplifier can also be vulnerable to temperature variations and create distortion. To get the best performance, careful biasing, the use of suitable load resistors and coupling capacitors are required.

Working of RC Coupled Amplifier

When the input voltage is applied to the base of the first transistor through an input capacitor C_{in} , it appears as the amplified form across its collector load R_C . The output of the

first stage is given to the base of the next stage through coupling capacitor CC. The second stage does the further amplification of the signal.



The **coupling capacitor** CC here helps in blocking the flow of DC components from the first stage to the second stage of the circuit. Thus the biasing of the second stage will not get disturbed. The coupling capacitor has some reactance and hence there would be some loss of signal voltage due to the drop across this capacitor. This loss varies inversely with the frequency of the applied input signal. As capacitive reactance is inversely proportional to the frequency of the signal.

$$X_C = 1/2\pi fC$$

We can notice that the total gain is less than the product of the gains from individual stages. This is because when the second stage is made to follow the first stage, the effective load resistance of the first stage is reduced due to the shunting effect of the input resistance of the second stage. This reduces the gain of the stage which is loaded by the next stage.

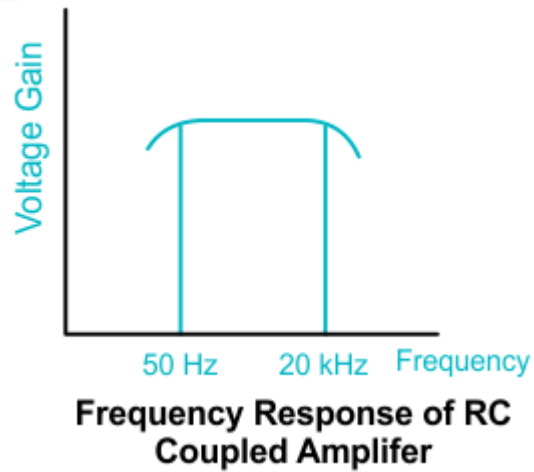
The ratio of output voltage VO to the input voltage VS is known as the **amplification or voltage gain** of the amplifier.

Also called the loop gain, denoted as AV

$$A_V = V_O/V_S$$

Frequency Response of RC Coupled Amplifier

Frequency response of RC coupled amplifier is uniform over the mid-frequency range of **50 Hz to 20 kHz**. The voltage gain drops off at low < **50 Hz** and high > **20 kHz** frequencies. The frequency response curve of RC coupled with the amplifier is shown in Figure. The behaviour of the amplifier is given below along with the bandwidth of an RC coupled amplifier between **50 Hz to 20kHz**.



Low Frequencies

At low frequencies ($< 50 \text{ Hz}$), the reactance of coupling capacitor CC is quite high and hence very small part of the signal will pass from one stage to the next stage. The reactance of CE is also high at low frequencies. This high reactance cannot shunt the emitter resistance R_E effectively. Due to these two factors, the voltage gain falls at low frequencies.

High Frequencies

At high frequencies ($> 20 \text{ kHz}$), the reactance CC is very small and it behaves as a short circuit. This increases the loading effect of the next stage and serves to reduce the voltage gain. Also, the capacitive reactance of the base-emitter junction is low, and at high frequencies increases the base current. Therefore the β value reduces. Due to these two reasons, the voltage gain falls at high frequencies.

Mid Frequencies

At mid frequencies (**50 Hz to 20 kHz**), the voltage gain of the amplifier is constant. The effect of coupling capacitors in this frequency range is such as maintaining a uniform voltage gain. Thus, as the frequency increases in this range, the reactance of CC decreases which tends to increase the gain. However, at the same time, lower reactance means higher loading of the first stage and hence lower gain. These two factors almost cancel each other, resulting in a uniform gain at mid-frequency.

Advantages of RC Coupled Amplifier

The advantages of RC Coupled Amplifier are as follows:

- The frequency response is excellent.
- This circuit is very compact and extremely light.
- Cost is low because it employs resistors and capacitors which are cheap.

Disadvantages of RC Coupled Amplifier

Disadvantages of RC Coupled Amplifier are as follows:

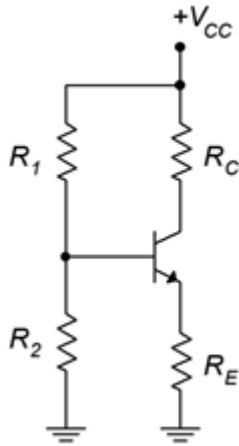
- Low voltage and power gain.
- They have the tendency to become noisy with age.

- Impedance matching is poor. It is because the output impedance of the RC coupled amplifier is several hundred ohms.

Applications of RC Coupled Amplifier

- The application of RC coupled amplifiers in music systems, has excellent audio fidelity over a wide range of frequencies.
- Therefore, they are widely used as voltage amplifiers (preamplifiers) **Ex:** in public address systems.

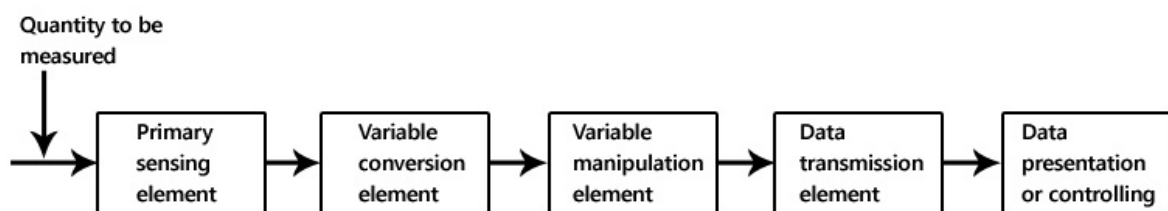
Concept of Voltage Divider Biasing



In this method, two resistors R_1 and R_2 are connected to the supply voltage V_{CC} and provide bias. Emitter resistor R_E provides stabilization. The name "voltage divider" comes from the voltage divider consisting of resistors R_1 and R_2 . The voltage drop across resistor R_2 will forward bias the base-emitter junction. Voltage divider biasing is known as a universal bias circuit one of the popular transistors that is biased with a voltage divider bias is a BJT transistor. The key advantage of a voltage divider bias is that it can incorporate more than one kind of voltage divider. On the other hand, the disadvantage of this bias is that signals tend to get mixed up at times.

Block diagram of Electronics Instrumentation System

It is a branch of engineering which deals with various types of instrument to record, monitor, indicate and control various physical parameters such as pressure, temperature, etc.



The block diagram shown above is of basic instrumentation system. It consist of primary sensing element, variable manipulation element, data transmission element and data presentation element.

Primary sensing element

The primary sensing element is also known as sensor. Basically transducers are used as a primary sensing element. Here, the physical quantity (such as temperature, pressure etc.) are sensed and then converted into analogues signal.

Variable conversion element

It converts the output of primary sensing element into suitable form without changing information. Basically these are secondary transducers.

Variable manipulation element

The output of transducer may be electrical signal i.e. voltage, current or other electrical parameter. Here, manipulation means change in numerical value of signal. This element is used to convert the signal into suitable range.

Data transmission element

Sometimes it is not possible to give direct read out of the quality at a particular place (Example – Measurement of temperature in the furnace). In such a case, the data should transfer from one place to another place through channel which is known as data transmission element. Typically transmission path are pneumatic pipe, electrical cable and radio links. When radio link is used, the electronic instrumentation system is called as telemetry system.

Data presentation or controlling element

Finally the output is recorded or given to the controller to perform action. It performs different functions like indicating, recording or controlling.

Other Explanation for Electronic Instrumentation System (Refer Separate Diagram from ECE-HoD)

- Fig shows block diagram of generalized electronics instrumentation system, in which number of element worked together to perform a desired function accurately i.e. to measure the measurand quantity and display it or record it.

Measurand (physical quantity or electrical quantity):

- The physical or electrical quantity which is to be measurand is called as measurand. If the input to the instrumentation is parameter like pressure, force, level, strain, displacement, temperature, flow, velocity etc. then these parameters is physical measurand. These measurand are applied to the transducer element.
- If the input is current, voltage and frequency then this parameter are called electrical measurand. These measurand are applied to the electrical conversion element.

Transducer and electrical conversion elements:

- If the measurand is physical quantity then it is converted into equivalent electrical signal with help of element which is called transducer. Transducer is a device which converts one form of energy into another. (Physical to Electrical).

- Most of the transducer are primary sensors which sense the measurand then convert it into electrical signal with the help of conversion element.
- If the measurand is already electrical signal like voltage, current or frequency then it is give to the electrical conversion element which convert the signal into more suitable form of signals such as 4-20mA, 1-5VDC, 1-10VDC etc.

Signal processing or signal conditioning:

- The output from the transducer elements is given to the signal processing or signal conditioning elements. Signal conditioning is a process to modify the output of transducer so that it can be measurand, controlled and acceptable by next stages.
- In electronic instrumentation system, filter, modulator, A/D converters, D/A converter, amplifiers, integrators, differentiators are the important signal conditioning circuits.
- This stage is required to convert the transducer output into an electrical quantity suitable for proper operation of the last stage or indicator.

Data transmission element:

- If the sensing element and data presentation element of the instrumentation system are away from each other (physically separated) in that case data transmission element is very important.
- This element provides a transmission path for the modified signals to travel from transducer element to the rest of instrumentation elements like recorders, controllers, displays etc.
- In electronic instrumentation system, typically the transmission path is a conducting lines (i.e. electrical cables). In electronic instrumentation system some time radio link is used as a transmission path then the system is called telemetry system.
- This element transmits the data to the remote located control room.

Data presentation element:

- The signal from the data lines are provided to the data presentation element. This element converts the signal into such from that it can be presented by some visual or audible means.
- Function performed by this stage may be demodulation, amplification, filtering, A/D conversion etc.
- This element modifies the signals in such a way that the signals are accepted by recorders, displays, indicators, printers, announcing systems etc. The out put of the data presentation element is provided to the recorders, controllers, and indicators as per the requirement of the user or operator or observer.

Output devices:

- The last stage of instrumentation system is required to provide the information about the measurand for immediate reorganization by the operator whatever output is presented on indicator.
- For example, if the system is simple instrumentation with display then measured parameter is displayed directly on the display. If it is required to produce alarm for the over range then indicator or buzzer is used. If the recorder like strip-chart or X-Y recorder etc. To record the data.
- If the system is control system in that case the measured data not only displayed or recorded but also compared with some reference value and control action is generated which is used to remove the error.

Power supply:

- This is a common unit for all instrumentation system. This provides power to all elements working in the instrumentation setup.
- If the transducer is active transducer (self-generating), in that case power supply is not required. For example if that transducer is thermocouple or piezoelectric crystal.
- For passive transducer power supply is required, for example if the transducer is thermistor or RTD, in that case power supply is important to the transducer block.

1. What is a rectifier?

A rectifier is an electronic device that converts an alternating current into a direct current by using one or more P-N junction diodes.

2. What is a half-wave rectifier?

The rectifier circuit that converts alternating current into the direct current is known as a half-wave rectifier circuit. The half-wave rectifier passes only one half of the input sine wave and rejects the other half.

3. Where is a half-wave rectifier used?

A half-wave rectifier is used in firing circuits and pulse generating circuits. Half-wave rectifiers are used along with step up and step down transformers to achieve the desired voltage.

4. How does a half-wave rectifier work?

Half wave rectifier circuit uses a PN junction to convert the supplied AC into DC. In a half-wave rectifier circuit, the load resistance is connected in series with the PN junction diode.

5. Is the half-wave rectifier better or the full-wave rectifier?

Full-wave rectifiers are more efficient than half-wave rectifiers. A full-wave rectifier passes twice as many waves as a half-wave rectifier. Hence, more of the input is transferred to the load.

6. What is the use of a filter in a half-wave rectifier?

The filter in a half-wave rectifier is used to smoothen the pulsating fluctuating DC component.

7. Define bridge rectifiers.

The bridge rectifier is a type of full-wave rectifier that uses four or more diodes in a bridge circuit configuration to convert alternating (AC) current to a direct (DC) current.

What is transistor biasing?

What is transistor biasing?

Transistor Biasing is the process of setting a transistors DC operating voltage or current conditions to the correct level so that any AC input signal can be amplified correctly by the transistor.